Riccardo Paoli

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Education

PhD **University of Pisa / CNR-ISTI**, Artificial Intelligence — Pisa, Italy
- Research Project: Neuro-symbolic reasoning and learning — Sept 2025 – present

MSc **University of Florence**, Theoretical Physics — Florence, Italy
- Thesis: Spectral Autoencoders: Systematic Analysis and Validation on Diffuse Optics Data — Sept 2022 – July 2025

Erasmus+ Exchange Program **École Normale Supérieure (ENS)**, Theoretical Physics — Paris, France
- Visiting student during the MSc program — Sept 2023 – Mar 2024

BSc **University of Florence**, Physics and Astrophysics — Florence, Italy
- Thesis: Stokes' Theorem in General Relativity — Sept 2019 – July 2022

Publications

MILP-SAT-GNN: Yet Another Neural SAT solver 2026
Franco Alberto Cardillo, Hamza Khyari, *Riccardo Paoli*, Umberto Straccia
10.1016/j.procs.2026.09.104 (Procedia Computer Science, Vol. 292, pp. 276–285 (KES 2026))

Spectral Autoencoders 2026
Riccardo Paoli, Lorenzo Chicchi, Duccio Fanelli, Lorenzo Buffoni
10.1109/COMPENG69955.2026.11766133 (2026 IEEE Workshop on Complexity in Engineering (COMPENG), pp. 1–5)

Learning-Based Reconstruction of Optical Properties in Bilayered Media from Single-distance Time-Resolved Reflectance Measurements 2026
Caterina Amendola, Giulia Maffeis, Lorenzo Buffoni, Lorenzo Chicchi, Francesco Coghi, Duccio Fanelli, Raffaele Marino, Fabrizio Martelli, *Riccardo Paoli*, Lorenzo Pattelli, Lorenzo Spinelli
arxiv.org/abs/2609.19786 (arXiv preprint arXiv:2609.19786)

Skills

Programming & Tools: Python, PyTorch, LaTeX, Git

AI & Computing: Graph Neural Networks (GNN), Neurosymbolic AI, SAT Solvers, Spectral Autoencoders

Languages

Italian: Native

English: Intermediate (B2)

Interests

Deep Learning, Neurosymbolic AI, General Relativity